

PRAMOD SAINI

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Education

Indian Institute Of Engineering Science And Technology, Shibpur

2022 – 2026

Bachelor of Technology in Electrical Engineering

Howrah, West Bengal

Subjects

Electrical Machines I / II / III, Power Systems I / II, Switchgear and Power System Protection, Power Electronics, Electric Drives, Renewable Energy, Control Systems I / II, Digital Electronics, Analog Electronics, Microprocessor and Microcontroller, Transducers and Instrumentation

Experience

Eastern Regional Load Despatch Centre (ERLDC), Grid India

June 2025 – July 2025

Intern

Kolkata, India

- Worked on “**Real-Time Fault Detection & Dynamic State Assessment using Multi-Model AI in WAMS**” for smart grids.
- Implemented **ML/DL models** for synchrophasor data analysis, improving fault detection accuracy and reliability.
- Enhanced fault detection speed, significantly reducing detection time for grid events.
- Gained hands-on exposure to **synchrophasor technology**, **AI/ML in power systems**, and **smart grid monitoring**.

Projects

Smart Visual Aid Device

Sep 2023 – Apr 2025

- Enhancing Mobility for the Visually Impaired (Embedded Systems, Sensors, IoT, Bluetooth)
- Developed a prototype assistive device to support visually impaired users in safe and independent navigation.
- Implemented **ultrasonic sensors** for real-time obstacle detection and distance measurement, and integrated **PIR sensors** for human motion and presence detection.
- Enabled **Bluetooth connectivity** for smartphone interaction and alerts, providing **audio/haptic feedback** to enhance user awareness and confidence.

Astable Multivibrator Using Operational Amplifier

Jan 2024 – Feb 2024

- Developed an op-amp-based astable multivibrator capable of generating stable clock pulses for use in timing and waveform circuits.
- Designed and implemented the circuit to produce continuous square wave signals; gained experience in analog simulation and hardware testing.

Technical Skills

Languages: SQL, MATLAB, 8085 Assembly Language

Software and Tools: Excel, MATLAB Simulink, Arduino UNO, VS Code, PSIM, LT Spice, Icarus Verilog, Yosys, OpenSTA, Magic Tool

Achievements

- **INSPIRE Award (Rajasthan Government):** Received for being among the top 10% of students in the Class 12 Board Examinations for outstanding academic performance under RBSC Board, Government of India.
- **Reliance Foundation Scholar:** Selected among the top 5,000 Indian undergraduates nationwide for academic excellence.

Extracurricular Activities

Club Member: EES – Electrical Engineers’ Society, IEST Shibpur

Volunteer: Sphuran 3.0 – Technical Fest of the Department of Electrical Engineering

Sports Engagements: Cricket, Table Tennis